

- 1 The lines l_1 and l_2 have equations $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} a \\ 2 \\ \frac{3}{2} \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 4 \\ b \end{pmatrix}$ where $\lambda, \mu \in \mathbb{R}$ and $a, b \in \mathbb{R}$.

(i) Given that l_1 and l_2 are parallel, write down the values of a and b . [2]

(ii) Find the shortest distance between l_1 and l_2 . [3]

(iii) Find a vector equation of the plane containing l_1 and l_2 . [1]

- 2 A sequence of positive real numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation $x_{n+1} = \sqrt{2x_n + 5}$ for $n \geq 1$.

(i) If this sequence converges to l , determine the exact value of l . [2]

(ii) By using a sketch of the graph of $y = \sqrt{2x + 5} - x$ and considering $x_{n+1} - x_n$, prove that

$$\begin{aligned} x_{n+1} &> x_n \text{ if } 0 < x_n < l, \\ x_{n+1} &< x_n \text{ if } x_n > l. \end{aligned} \quad [3]$$

(iii) Describe the sequence when $x_1 = l$. [1]

- 3 (i) Sketch the graph of $y = x - \ln x$. By adding a suitable graph to the sketch, find the set of values of x that satisfies $x^3 - x^2 \ln x - 5x^2 + 1 \leq 0$. [4]

(ii) Using an algebraic method, solve the inequality $\frac{18}{(1-x)^2} \leq \frac{x+6}{x-1}$. [3]

- 4 Given that $z_1 = \sqrt{3} + i$ and $z_2 = 1 - \sqrt{3}i$, find $|z_1 - z_2|$ and $\arg(z_1 - z_2)$. [2]

The complex number z satisfies the equation $|z - z_1| = |z_2|$. Sketch, on an Argand diagram, the locus of point P representing the complex number z . [2]

Hence, find the least possible value of $\arg(z - z_2)$. [3]

- 5 A certain supermarket sells only three brands of health drinks namely Mud, Soil and Rock which are priced at \$2.00, \$1.80 and \$1.50 per bottle respectively.

Discounts are offered for each brand as follows:

20% discount when more than 80 bottles of Mud are bought;
15% discount when more than 50 bottles of Soil are bought;
10% discount when more than 30 bottles of Rock are bought.

Andy orders 200 bottles of health drinks from the supermarket for a company function. As the preference for each brand differs among the colleagues in the company, he has to order at least 60 bottles of each of the 3 brands.

Andy spent \$323.05 after an overall discount of \$27.45.

Find the number of bottles of each brand Andy bought from the supermarket, explaining your answer clearly. [7]

- 6 The functions f and g are defined as follow:

$$f: x \mapsto -x^2, \quad x \in \mathbb{R}, \quad -1 \leq x \leq 3,$$

$$g: x \mapsto e^{-\sqrt{9-x}}, \quad x \in \mathbb{R}, \quad x \leq 0.$$

Explain why gf exists. [2]

Express gf as a function in a similar manner and find its range in exact form. [4]

Determine whether gf has an inverse. [2]

- 7 Given that $U_r = \frac{1}{\sqrt{r} + \sqrt{r-1}}$, show that $\sum_{r=1}^n U_r = \sqrt{n}$

(i) by using the method of differences, [3]

(ii) by mathematical induction. [4]

Hence, prove that $\sum_{r=1}^n \frac{1}{2\sqrt{r}} < \sqrt{n}$. [2]

- 8 Given that $y = \sin(\ln(1+2x))$, prove that $(1+2x)^2 \frac{d^2y}{dx^2} + 2(1+2x) \frac{dy}{dx} + 4y = 0$.

Find the Maclaurin's series for $\sin(\ln(1+2x))$ up to and including the term in x^3 . [7]

Hence, show that $\cos(\ln(1+2x)) \approx 1 - 2x^2$. [2]

- 9 By using the substitution $2x = \sin \theta$, show that

$$\int_0^k \frac{x^2}{\sqrt{1-4x^2}} dx = \frac{1}{16} \left(\sin^{-1} 2k - 2k\sqrt{1-4k^2} \right), \quad \text{where } k \in \mathbb{R}^+. \quad [5]$$

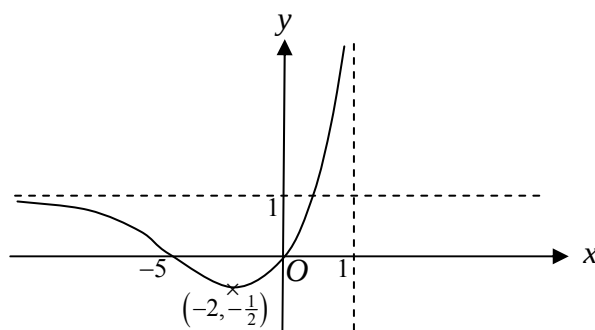
Hence, find the value of k , correct to 3 significant figures, such that

$$\int_0^k x \sin^{-1} 2x \, dx = \int_0^{0.3} \sqrt{x \sin^{-1} 2x} \, dx. \quad [4]$$

- 10 (a) Find all the asymptotes of $y = \frac{x^2 + 2x}{(x + \lambda)(x - 1)}$, where $\lambda < -1$.

Hence, sketch the curve. (*Coordinates of the turning points are not required.*) [4]

(b)



The diagram shows a sketch of the graph of $y = f(x)$.

Sketch, on separate clearly labeled diagrams, the graphs of

(i) $y = \frac{1}{f(x)},$ [3]

(ii) $y = f'(x).$ [3]

- 11 (a) The terms $u_1, u_2, u_3, \dots$ form an arithmetic sequence with first term a and having non-zero common difference d .

(i) Given that the sum of the first 8 terms of the sequence is 98 more than u_{29} , find the first term of the sequence. [2]

(ii) If u_{15} is the first term in the sequence greater than 196, show that $13 < d \leq 14$. [3]

- (b) The terms $v_1, v_2, v_3, \dots$ form a geometric sequence with common ratio r . Another sequence $\{w_n\}$ is then defined by $w_n = v_{2n-1} + v_{2n}$ for all positive integers n .

(i) Show that w_n is a geometric sequence with common ratio r^2 . [2]

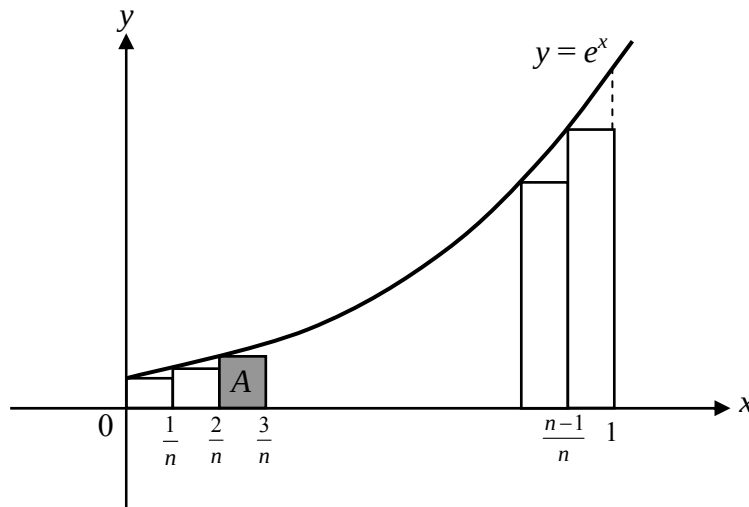
(ii) Given that $v_1 = 4$ and the sum of all the odd-numbered terms $w_1, w_3, w_5, \dots$ in the sequence $\{w_n\}$ is $\frac{32}{15}$, find the value of r and $\sum_{r=1}^{\infty} v_r$. [4]

- 12 (a) A curve C is defined parametrically by the equations,

$$x = \frac{t}{t-1}, \quad y = \frac{t^2(t-1)^2}{t^3-1}, \quad \text{where } t \neq 1.$$

Find the exact area bounded by the curve C , the x -axis and the line $x = \frac{1}{3}$. [5]

- (b) The graph $y = e^x$, for $0 \leq x \leq 1$ is shown in the diagram below. The region bounded by the curve, the axes and the line $x = 1$, is rotated through four right angles about the x -axis to form a solid.



- (i) The shaded region A forms a circular disc when it is rotated through four right angles about the x -axis. Show that the volume of this disc is $\frac{\pi}{n} e^{\frac{4}{n}}$ unit³. [2]
- (ii) Circular discs, each of width $\frac{1}{n}$, are used to approximate the volume of the solid formed above. By considering the volume of the discs, show that for all $n \in \mathbb{N}^+$,

$$\frac{\pi}{n} \left(\frac{1-e^2}{1-e^{\frac{2}{n}}} \right) < \frac{\pi}{2} (e^2 - 1) \quad [4]$$

End of Paper